6 A cylinder that contains a fixed amount of an ideal gas is shown in Fig. 6.1.

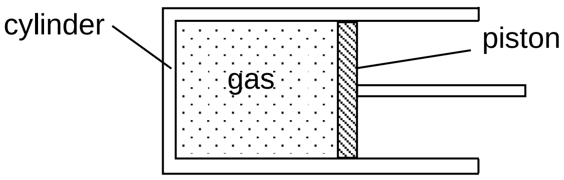

Fig. 6.1

The cylinder is fitted with a piston that moves freely.

(a) Use the kinetic theory of gases to explain

(i) the origin of the pressure of the gas in the cylinder,

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.... [3]

L2	**Gas atoms are in constant random motions**, and they continually collide with the walls of the cylinder. **When a gas atom collides with the wall of the cylinder, it rebounds and its velocity changes direction, hence there is a change in momentum of the gas atom.**	B1
	By Newton's second law of motion, there is a resultant force exerted by the wall on the gas atom which is proportional to the rate of change in momentum.	B1
	By Newton's third law, the gas atom exerts a force of equal magnitude and opposite direction on the wall. **The pressure exerted by the gas is the total force per unit area of the container walls exerted by all the gas atoms on the walls of its container**.	B1
	***An answer referring to kinetic theory of gases (i.e. referring to the movement of the gas atoms) is expected. No mark awarded if candidate explains with reference to macroscopic properties.*	

(ii) why the mean velocity of the atoms of the gas is zero.

....

[Turn over

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…………………………………………………………………………………………………… [2]

L2 The **gas atoms are in random motion** and this means that **every atom has equal probability of moving in every direction.**
Also, the mean speed in all directions are equal. **B1**

Consequently, the **vector sum of all their velocities is equal to zero**. Hence, the mean velocity is zero. **B1**

(b) Fig. 6.2 shows the variation of pressure and volume of the monoatomic ideal gas in the cylinder. The gas is initially at state W.

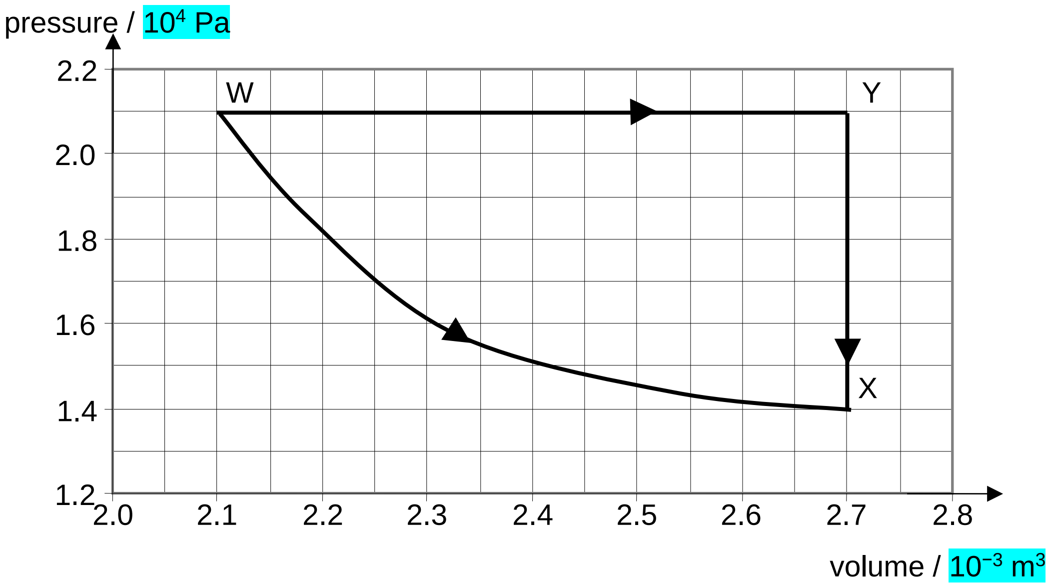

Fig. 6.2

(i) Determine the change in internal energy of the gas when it is taken from state W to state X along the curved path.

change in internal energy =……………………………… J [2]

L2 For a monoatomic ideal gas,

$$\Delta U_{wx} = \frac{3}{2}nR\Delta T = \frac{3}{2}\Delta(pV)$$

$$\Delta U_{wx} = \frac{3}{2}(p_x V_x - p_w V_w)$$

$$\Delta U_{wx} = \frac{3}{2}\left[(1.4x10^4)(2.7x10^{-3}) - (2.1x10^4)(2.1x10^{-3})\right]$$ **M1**

$$\Delta U_{wx} = -9.45 \text{ J}$$ **A1**

(ii) The same resultant change in state of the gas may be achieved by stages WY and YX.

Determine the net heat supplied to the gas during the change from W to Y to X.

			heat supplied = …………………………… J	[4]
		L2	**The change in internal energy for both paths, W to X or W to Y to X are the same.** $\Delta U_{WX} = \Delta U_{WYX} = -9.45\ \text{J}$ Using Fig. 6.2, work done by the gas $= (2.1 \times 10^4)(2.7 - 2.1) \times 10^{-3} = 12.6\ \text{J}$ Using first law of thermodynamics, $\Delta U_{WYX} = Q + W_{WYX}$ $-9.45 = Q + (-12.6)$ $Q = +3.15\ \text{J}$	**C1** **M1** **M1** **A1**

[Total: 11]

7	(a)	Fig. 7.1 shows the variation with potential difference V of the current I for a filament lamp X